

## Project Initialization and Planning Phase

|               |                                          |
|---------------|------------------------------------------|
| Date          | 30 June 2026                             |
| Team ID       |                                          |
| Project Title | EduGenie – AI Powered Learning Assistant |
| Maximum Marks | 5 Marks                                  |

### Project Proposal (Proposed Solution) report

The proposal report aims to enhance education through Artificial Intelligence by providing personalized learning support, instant doubt resolution, concept explanations, quiz generation, and content summarization. The solution improves learning efficiency, engagement, and academic performance using Google Gemini AI.

| Project Overview  |                                                                                                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective of EduGenie is to provide an AI-powered learning assistant that helps students understand concepts, generate quizzes, summarize content, and receive personalized learning support.                                 |
| Scope             | The project focuses on enhancing the learning experience through artificial intelligence by providing instant doubt resolution, concept explanations, quiz generation, and personalized study assistance.                                 |
| Problem Statement |                                                                                                                                                                                                                                           |
| Description       | Students often face difficulties in understanding complex concepts, finding relevant learning resources, and receiving personalized guidance. Traditional learning methods are time-consuming and may not meet individual learning needs. |
| Impact            | These challenges can reduce learning efficiency, increase frustration, and negatively affect academic performance. Addressing these issues can improve learning outcomes and student engagement.                                          |
| Proposed Solution |                                                                                                                                                                                                                                           |
| Approach          | The solution uses Google Gemini AI and FastAPI to deliver intelligent educational assistance through concept explanations, quiz generation, text summarization, and personalized learning recommendations.                                |
| Key Features      | AI-powered Question Answering<br>Concept Explanation, Quiz Generation, Text Summarization, Personalized Learning Path                                                                                                                     |

## Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                     |
|-------------------------|-----------------------------------------|----------------------------------------------|
| <b>Hardware</b>         |                                         |                                              |
| Computing Resources     | CPU/GPU specifications, number of cores | Intel/AMD Processor, Multi-Core CPU          |
| Memory                  | RAM specifications                      | 8 GB                                         |
| Storage                 | Disk space for data, models, and logs   | 1 TB SSD                                     |
| <b>Software</b>         |                                         |                                              |
| Frameworks              | Python frameworks                       | FastAPI                                      |
| Libraries               | Additional libraries                    | google-generativeai, requests, python-dotenv |
| Development Environment | IDE                                     | HTML, CSS, JavaScript                        |
| <b>Data</b>             |                                         |                                              |
| Data                    | Source, size, format                    | Google Gemini API                            |